



Alexey Mazurenko

AI ENGINEER

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HIGHLIGHTS

- ~2,000 docs · ~200 questions/day · Recall@1 87% held-out
- Report preparation: 2 hours → 2 minutes
- ~85% normalized SQL result correctness (held-out)
- Open-source demos — RAG Chat and AI Data Pilot on GitHub

PROFILE

AI engineer with 7+ years in software engineering, including **2+ years building production LLM systems** at MTS Web Services (MWS AI) — the AI division of MTS, one of Russia's largest telecom operators (~80M+ subscribers). Shipped two internal products end-to-end: a document RAG assistant (**~2,000 docs; held-out Recall@1 53% → 87%**) and a multi-agent analytics platform (**~85% normalized SQL result correctness**) that **cut report preparation from 2 hours to 2 minutes**.

Internal production at MTS (~12 teams, ~200 RAG questions/day); the public site and GitHub demos use a test corpus. RAG production default is **GLM-5.3-flash** (TTFT ~2.5–3 s); OpenAI, Anthropic and Ollama are interchangeable.

Own the path from prototype to production: Python / FastAPI, hybrid retrieval, held-out evaluation, React / Next.js, Docker / Kubernetes / CI/CD.

More about my projects and experience — on my website: lyakoway.vercel.app

EXPERIENCE

AI Engineer

MTS Web Services (MWS AI)

Apr 2024 — present · Moscow

PROJECTS

RAG Chat — AI assistant with Retrieval-Augmented Generation

An AI assistant for working with documents — uploads PDF, Word and Excel files, answers questions about their content and provides links to the source pages and document fragments.

- **RAG pipeline:** document indexing → chunking (tiktoken) → embeddings (fastembed) → hybrid BM25 + vector retrieval (RRF) → LLM answer generation → source citation.
- **Three modes:** RAG Chat → AI Agent → Vector Search — switching within one application.
- **Architecture:** Python / FastAPI → ChromaDB → fastembed → GLM-5.3-flash default (OpenAI / Anthropic / Ollama) → SSE → React / TypeScript; SQLAlchemy — conversation history.
- **Evaluation:** held-out set (~180 queries): hybrid BM25 + RRF raised Recall@1 from 53% to 87%; cross-encoder reranker tested and rejected (~45 p.p., +~3 s).
- **Quality:** 63 pytest tests + CI; TTFT ~2.5–3 s on GLM-5.3-flash.

AI Data Pilot — multi-agent analytics platform

Automates the path from a user's question to a ready analytical result.

- **Data Agent:** turns natural-language questions into SQL, runs multi-step data analysis, detects trends and deviations, and produces tables, charts and analytical conclusions.
- **Knowledge Agent (RAG):** answers questions over internal technical documentation and uploaded files (PDF, Word, Excel), grounded in the retrieved sources.
- **Architecture:** Python / FastAPI → SQLAlchemy → Agent Loop (ReAct) → Tool Calling → Text-to-SQL → SQL guard → PostgreSQL / ClickHouse → analytics layer → SSE → React / TypeScript; the Knowledge Agent's RAG core — hybrid retrieval BM25 + vector embeddings (fastembed) → LLM → source citation.
- **Quality:** 174 pytest tests + CI; held-out SQL eval — ~98% execution, ~85% normalized result correctness; self-correction on failed queries (GLM-4.6).

CORE TASKS

- Designed and shipped production AI agents — tool calling, orchestration, error handling and recovery.
- Built and measured RAG pipelines — hybrid BM25 + vector search, source citations; rejected a slower reranker on data.
- Built held-out evaluation suites for regression checks of retrieval, prompts and models.
- Implemented Text-to-SQL with SQL guardrails and self-correction.
- Owned AI platform services end-to-end — backend (Python / FastAPI), frontend (React / Next.js), infrastructure (Docker / Kubernetes / CI/CD).

KEY RESULTS

RAG Chat

- Internal MTS production: ~2,000 documents (~20k chunks), ~12 teams, ~200 questions/day. Public demo = test corpus.
- Held-out Recall@1 53% → 87% after hybrid BM25 + RRF; answers cite the source fragment.
- Cuts lookup from minutes to seconds on regulations, contracts and HR policies.

SKILLS

LLM & AI Agents

- LLM APIs (OpenAI / Anthropic / GLM-5.3-flash / Ollama)
- AI AgentsAgent Loop (ReAct)
- Tool CallingText-to-SQL
- Multi-agent router (2 specialists)
- Prompt Engineering
- Latency & Cost Tuning (TTFT, \$ per query)

RAG & Retrieval

- RAGHybrid Search (BM25 + Vector)
- RRFChunking (tiktoken)fastembed
- ChromaDBSource Citations

Evaluation & Quality

- Held-out Golden SetsRecall@K / MRR
- LLM-as-a-Judge
- Grounding / Anti-Hallucination
- Prompt EvaluationRegression Testing

AI Data & Backend

- PythonFastAPISQLAlchemy
- PostgreSQLClickHouse
- SQL Guard (read-only, timeouts)SSE

Frontend & Infrastructure

- React / Next.jsTypeScriptDocker
- KubernetesCI/CDGit

EDUCATION

Master's degree — Applied Mathematics
Moscow State University of Civil Engineering (MGSU), 2014

KEY RESULTS

AI Data Pilot

- Internal MTS production: ~15 analysts, ~80 reporting scenarios/week over PostgreSQL, ClickHouse and Excel. Public demo = test data.
- Held-out SQL: ~85% normalized result correctness; report prep 2 hours → 2 minutes — the agent writes, executes and repairs SQL.
- One interface over PostgreSQL, ClickHouse and Excel, with the SQL dialect adapted automatically.

STACK

- RAG Chat

PythonFastAPISQLAlchemyRAGAI AgentsLLM APIChromaDBfastembed

OllamaSSEReactTypeScriptVite
- AI Data Pilot

PythonFastAPISQLAlchemyText-to-SQLAgent Loop (ReAct)Tool CallingRAG

BM25 + Vector SearchfastembedPostgreSQLClickHouseSSEReact 19

TypeScriptpytest

DEMO

- RAG Chat — case study — [lyakoway.vercel.app/portfolio/rag-chat](#)
- RAG Chat — code — [github.com/lyakoway/ai-RAG-chat](#)
- AI Data Pilot — case study — [lyakoway.vercel.app/portfolio/ai-data-pilot](#)
- AI Data Pilot — code — [github.com/lyakoway/ai-data-pilot](#)

EXPERIENCE (CONTINUED)

Senior Frontend Developer
MTS Web Services

Feb 2019 — Apr 2024 · Moscow

PROJECTS

MTS Profile

A module for storing and visualizing customer data with access management across ecosystem products.

Ecosystem widgets

An embeddable navigation and personalization module for b2c/b2b products.

KEY RESULTS

- Delivered a profile ownership transfer model and a linked-accounts management model.
- Automated user data verification via Gosuslugi with biometrics.
- Delivered the full access management and authorization cycle — access recovery, sign-in and an authentication-method change history.
- Integrated bank card payments and ecosystem widgets into the company's digital products.

STACK

- ReactNext.jsTypeScriptRedux ToolkitSvelteStyled-ComponentsWebpackJestNode.js
- Express